

What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence (AI) in which machines learn patterns and relationships from data and use that learning to make predictions or decisions on new, similar data.

In simple words, machines learn from existing data and then use that knowledge to make predictions on unseen data without being explicitly programmed for every situation.

Machine Learning can be broadly divided into three main types:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Some common tasks in Machine Learning include anomaly detection and dimensionality reduction.

Machine Learning Pipeline and EDA

A Machine Learning pipeline consists of several steps, and one of the most important steps is Exploratory Data Analysis (EDA).

EDA is the process of understanding, analyzing, and preparing data before feeding it into Machine Learning or Deep Learning models. During this phase, we visualize the data and perform preprocessing tasks such as data cleaning, handling missing values, data transformation, and data wrangling.

Different types of data require different preprocessing techniques, so it is important to understand the various types of data used in Machine Learning.

Different Types of Data

Data can be categorized based on its structure, representation, and labeling.

Based on Structure

Structured Data is organized in a predefined format, usually in rows and columns. It is easy to store, query, and analyze.

Examples:

- Sales records
- Student databases
- Employee records

Unstructured Data does not follow a predefined format or structure.

Examples:

- Text documents
- Images
- Audio files
- Videos

Based on Representation

Numerical Data consists of measurable quantities represented by numbers, such as age, salary, height, and weight.

Numerical data can be of two types:

- **Discrete Data:** Countable values, such as the number of students in a class or the number of cars owned.
- **Continuous Data:** Values that can take any value within a range, such as height, weight, and temperature.

Categorical Data represents labels or categories rather than numerical values.

Examples:

- Gender
- Blood Group
- City

Categorical data can be further divided into:

- **Nominal Data:** Categories with no meaningful order, such as hair color or blood group.
- **Ordinal Data:** Categories with a meaningful order, such as clothing sizes (Small, Medium, Large) or customer satisfaction ratings.

Based on Labeling

Labeled Data contains input features along with their corresponding target outputs (labels). It is mainly used in supervised learning.

For example, a dataset containing house features and their corresponding prices.

Unlabeled Data contains only input features and no target labels. It is commonly used in unsupervised learning techniques such as clustering.

For example, a collection of images without any annotations or categories.

Specialized Types of Data

Time Series Data consists of observations collected over time at regular intervals. It is commonly used to analyze trends and make future predictions.

Examples:

- Stock prices
- Weather data
- Daily sales records

Text Data consists of words, sentences, and documents. Since Machine Learning algorithms work with numbers, text data is usually converted into numerical representations before training a model.

Applications include:

- Sentiment Analysis
- Document Classification
- Spam Detection
- Chatbots

In summary, Machine Learning is all about learning patterns from data and using those patterns to make predictions or gain insights. Understanding different types of data is essential because the way we preprocess and analyze data depends heavily on its type.

EDA Analysis on Titanic Dataset

EDA on Titanic Dataset

```
In [3]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [4]: df = pd.read_csv("Titanic-Dataset.csv")
```

```
In [6]: df.head()
```

```
Out[6]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0

```
In [11]: df.describe()
```

Out[11]:

	PassengerId	Survived	Pclass	Age	SibSp	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.0
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.3
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.8
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.0
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.0
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.0
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.0
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.0

In [13]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null    int64
1   Survived        891 non-null    int64
2   Pclass          891 non-null    int64
3   Name            891 non-null    object
4   Sex             891 non-null    object
5   Age             714 non-null    float64
6   SibSp           891 non-null    int64
7   Parch           891 non-null    int64
8   Ticket          891 non-null    object
9   Fare            891 non-null    float64
10  Cabin           204 non-null    object
11  Embarked        889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [8]: `df.shape`

Out[8]: (891, 12)

In [65]: `df.columns`

Out[65]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'], dtype='object')

In [66]: `df.isnull().sum()`

```
Out[66]: PassengerId      0
         Survived        0
         Pclass          0
         Name            0
         Sex             0
         Age            177
         SibSp           0
         Parch           0
         Ticket          0
         Fare            0
         Cabin          687
         Embarked        2
         dtype: int64
```

```
In [75]: df["Age"] = df["Age"].fillna(df["Age"].mean())
```

```
In [76]: df.isnull().sum()
```

```
Out[76]: PassengerId      0
         Survived        0
         Pclass          0
         Name            0
         Sex             0
         Age            0
         SibSp           0
         Parch           0
         Ticket          0
         Fare            0
         Cabin          687
         Embarked        2
         dtype: int64
```

```
In [77]: df["Cabin"] = df["Cabin"].fillna(df["Cabin"].mode())
```

```
In [78]: df.isnull().sum()
```

```
Out[78]: PassengerId      0
         Survived        0
         Pclass          0
         Name            0
         Sex             0
         Age            0
         SibSp           0
         Parch           0
         Ticket          0
         Fare            0
         Cabin          685
         Embarked        2
         dtype: int64
```

```
In [79]: df = df.drop(columns = ["Cabin"])
```

```
In [80]: df.isnull().sum()
```

```
Out[80]: PassengerId    0
         Survived      0
         Pclass       0
         Name         0
         Sex          0
         Age          0
         SibSp        0
         Parch        0
         Ticket       0
         Fare         0
         Embarked     2
         dtype: int64
```

```
In [81]: df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode())
```

```
In [82]: df.isnull().sum()
```

```
Out[82]: PassengerId    0
         Survived      0
         Pclass       0
         Name         0
         Sex          0
         Age          0
         SibSp        0
         Parch        0
         Ticket       0
         Fare         0
         Embarked     2
         dtype: int64
```

```
In [84]: plt.figure(figsize=(6,4))

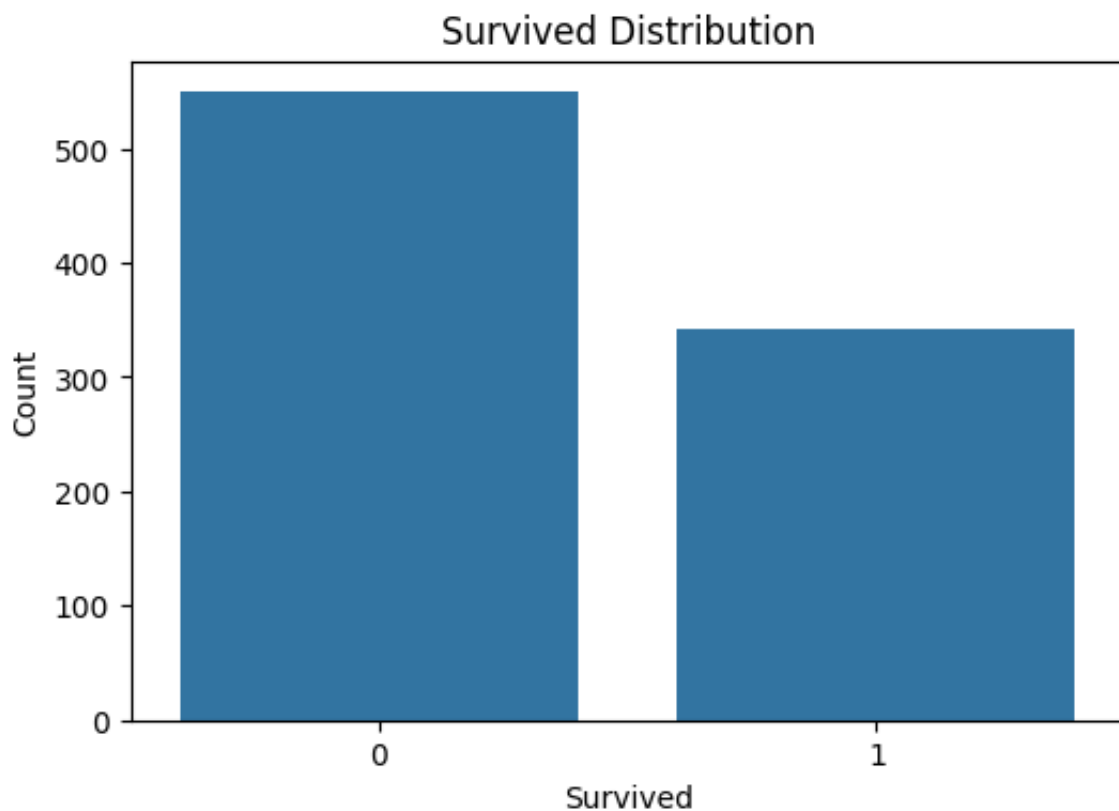
         sns.countplot(x='Survived', data=df)

         plt.title("Survived Distribution")

         plt.xlabel("Survived")

         plt.ylabel("Count")

         plt.show()
```



```
In [21]: numerical_cols = df["PassengerId"], df["Survived"], df["Pclass"], d
```

```
In [22]: print(numerical_cols)
```

```
(0      1
1      2
2      3
3      4
4      5
...
886    887
887    888
888    889
889    890
890    891
Name: PassengerId, Length: 891, dtype: int64, 0      0
1      1
2      1
3      1
4      0
..
886    0
887    1
888    0
889    1
890    0
Name: Survived, Length: 891, dtype: int64, 0      3
1      1
2      3
3      1
4      3
..
```



```

886     2
887     1
888     3
889     1
890     3
Name: Pclass, Length: 891, dtype: int64, 0      22.0
1      38.0
2      26.0
3      35.0
4      35.0
...
886     27.0
887     19.0
888      NaN
889     26.0
890     32.0
Name: Age, Length: 891, dtype: float64, 0      1
1      1
2      0
3      1
4      0
..
886     0
887     0
888     1
889     0
890     0
Name: SibSp, Length: 891, dtype: int64, 0      0
1      0
2      0
3      0
4      0
..
886     0
887     0
888     2
889     0
890     0
Name: Parch, Length: 891, dtype: int64, 0      7.2500
1      71.2833
2      7.9250
3      53.1000
4      8.0500
...
886     13.0000
887     30.0000
888     23.4500
889     30.0000
890     7.7500
Name: Fare, Length: 891, dtype: float64)

```

```

In [68]: plt.figure(figsize=(7,4))

sns.histplot(

    df["Age"],

```

```

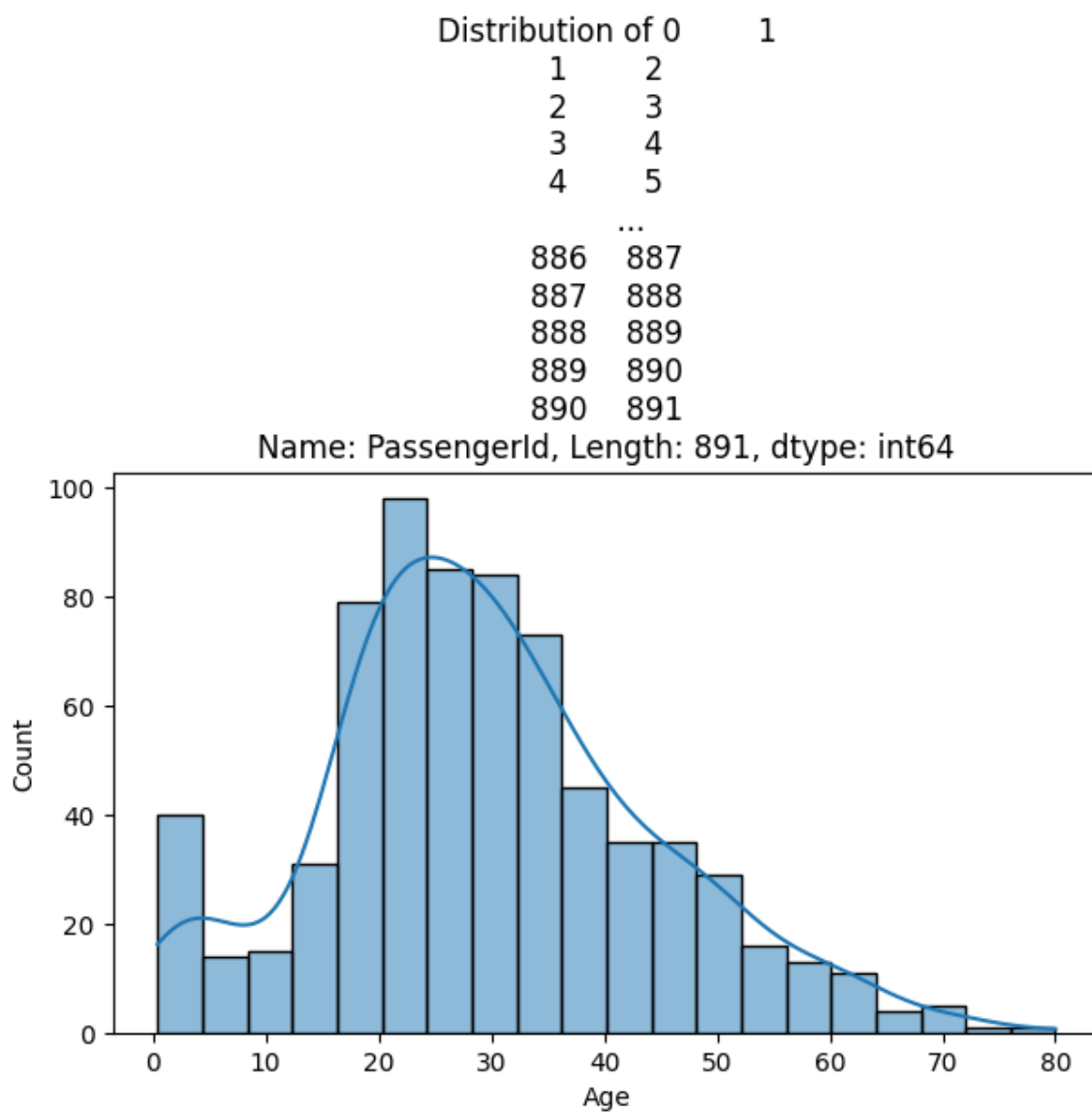
    kde=True,

    bins=20
)

plt.title(f"Distribution of {col}")

plt.show()

```



```

In [64]: plt.figure(figsize=(16,10))

sns.boxplot(

    x='Pclass',

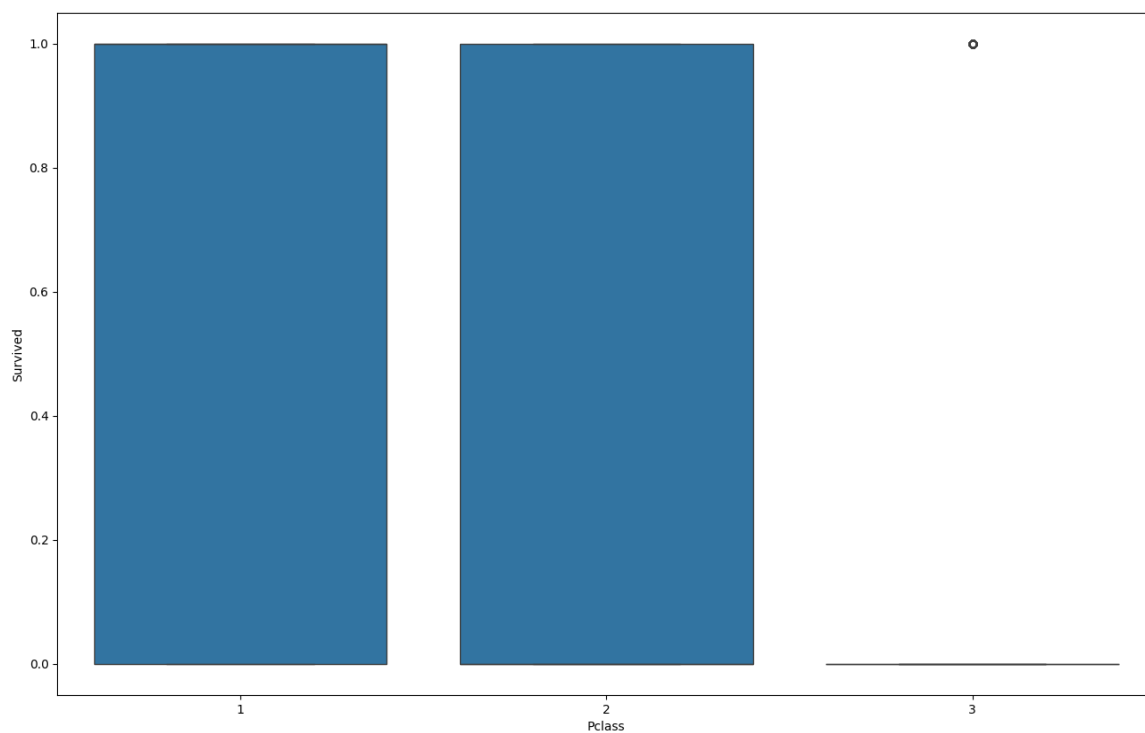
    y="Survived",

    data=df

)

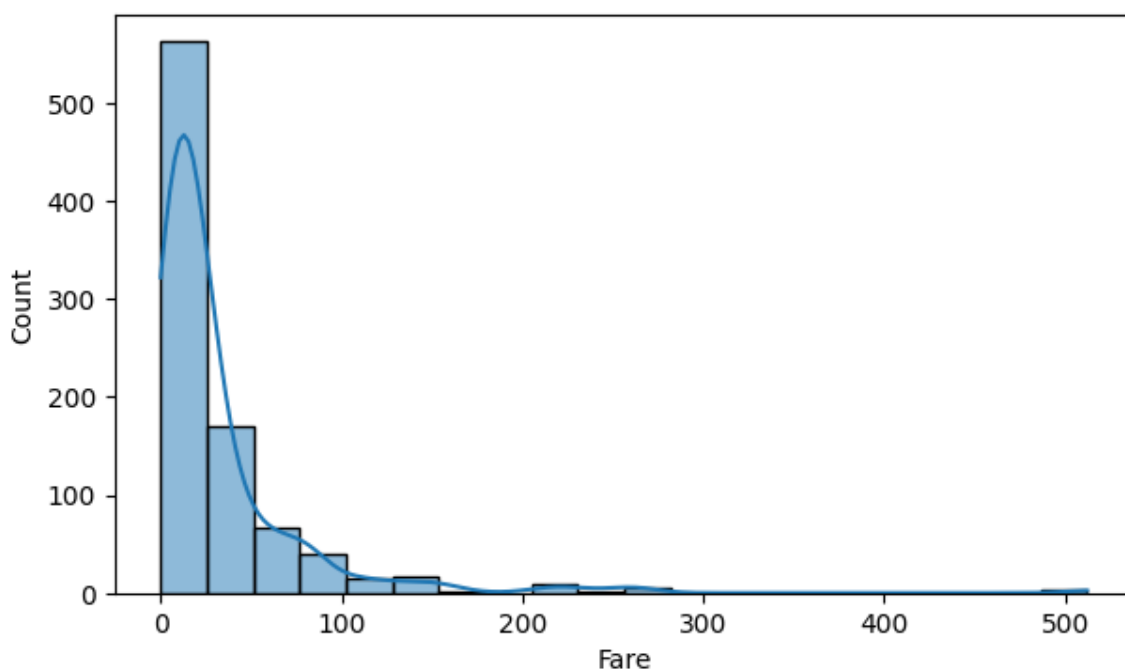
```

```
plt.show()
```



```
In [113... plt.figure(figsize=(7,4))  
sns.histplot(  
    df["Fare"],  
    kde=True,  
    bins=20  
)
```

```
Out[113... <Axes: xlabel='Fare', ylabel='Count'>
```



```
In [103... df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 891 entries, 0 to 890  
Data columns (total 8 columns):  
#   Column      Non-Null Count  Dtype  
---  -  
0   PassengerId  891 non-null    int64  
1   Survived     891 non-null    int64  
2   Pclass       891 non-null    int64  
3   Age          891 non-null    float64  
4   SibSp        891 non-null    int64  
5   Parch        891 non-null    int64  
6   Fare         891 non-null    float64  
7   Embarked     889 non-null    object  
dtypes: float64(2), int64(5), object(1)  
memory usage: 55.8+ KB
```

```
In [104... df = df.drop(columns=['Embarked'])
```

```
In [105... df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass           891 non-null    int64
3   Age              891 non-null    float64
4   SibSp            891 non-null    int64
5   Parch            891 non-null    int64
6   Fare             891 non-null    float64
dtypes: float64(2), int64(5)
memory usage: 48.9 KB
```

```
In [109... plt.figure(figsize=(8,6))

sns.heatmap(

    df.corr(),

    annot=True,

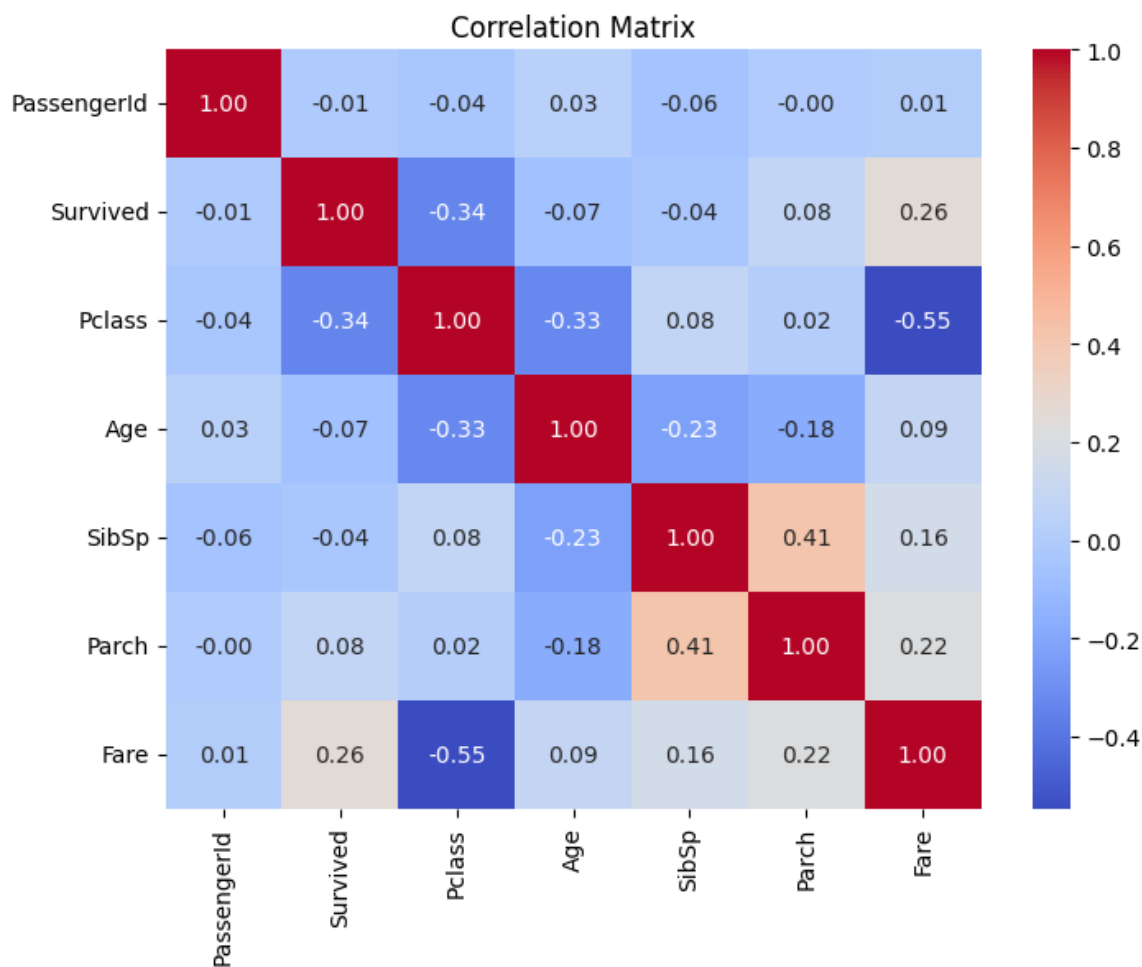
    cmap='coolwarm',

    fmt='.2f'

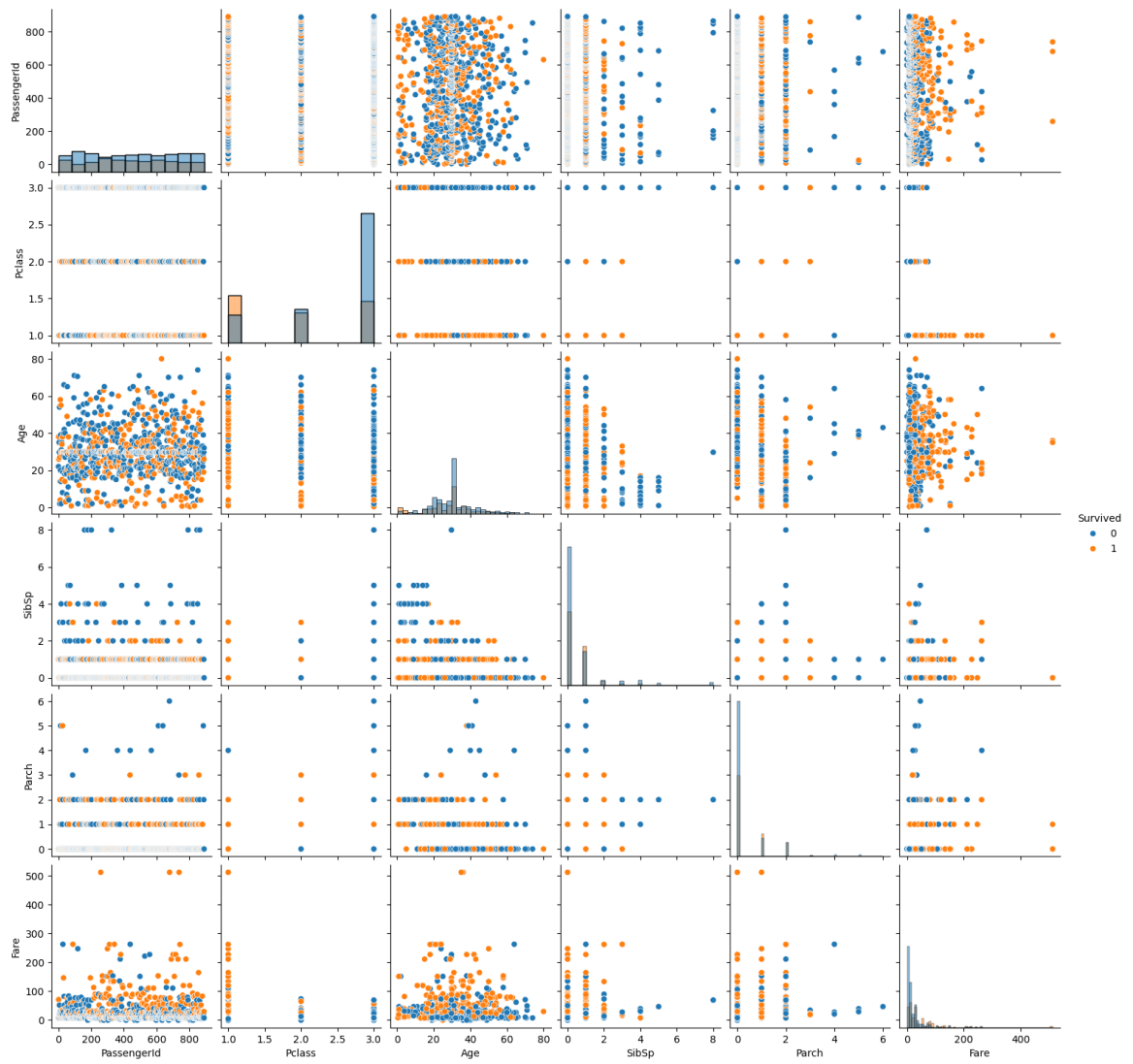
)

plt.title("Correlation Matrix")

plt.show()
```



```
In [112... sns.pairplot(
    df,
    hue='Survived',
    diag_kind='hist'
)
plt.show()
```



In []: